Euiseok Han

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Research Interests

•Uncertainty-Aware Decision Making for Autonomous Systems: Multimodal Perception, State Estimation, Robot Policies, 3D Scene Understanding

Education

•Korea Advanced Institute of Science and Technology, Daejeon *Mar. 2020 – Aug. 2027 (Expected)*
B.S. in Electrical Engineering and Computer Science (Double Major) GPA: 4.02/4.3

– *Coursework*: Machine Learning for 3D Data, Kalman Filtering (graduate), Mathematical Physics

•University of Texas at Austin *Aug. 2025 – Dec. 2025*
Exchange Program GPA: 4.0/4.0

Publications

[1] **SenseFuse: Label-Free Fusion of Image and Shape Encoders for Open-Vocabulary 3D Instance Segmentation** *Under review*
Euiseok Han, Tri Ton, Hwanhee Kim, Seungyeon Ryu, Chang D. Yoo
ICRA 2027 – IEEE International Conference on Robotics and Automation [arXiv:2609.20475] [code]

[2] **Reference-Filter-Driven Transition Probabilities for IMM-Based Satellite Maneuver Detection** *Preprint*
Euiseok Han, Sangheon Choi, Seung-Hyun Kong
arXiv preprint [arXiv:2610.11242] [interactive demo]

Research Experience

•Artificial Intelligence and Machine Learning Lab, KAIST *Jan. 2025 – Feb. 2025, Jan. 2026 – Present*
Undergraduate Researcher | Advisor: Prof. Chang D. Yoo Daejeon, KR

– URP: *From Point Cloud to Actions: 3D Structural Keypoint-Conditioned Hierarchical Robot Policy Learning*
– Revealed that frozen 2D image and 3D shape encoders fail differently and are weakly correlated (Spearman ρ 0.08–0.23, 5.9% error overlap), versus 0.68 and 41.4% for a 2D–2D pair [1]
– Devised a 2D-3D fusion framework with a closed-form, label-free scene-level weight from sensitivity maximization (3.5 ms), achieving a median gain of +7.1 points and 93% of the oracle-weight gain
– Built UR5 Gymnasium and joystick teleoperation interfaces; trained a diffusion policy on the collected demos

•NC Lab, KAIST *Mar. 2025 – Aug. 2025*
Undergraduate Researcher | Advisor: Prof. Junehwa Song Daejeon, KR

– *Socially-appropriate Use of Noise-making Everyday Appliances in High-density Urban Housing*
– Participated in experiment design and conducted a preliminary experiment using Zigbee modules

•Control and Intelligent Systems Lab, KAIST *Jul. 2024 – Aug. 2024*
Undergraduate Researcher | Advisor: Prof. Heejin Ahn Daejeon, KR

– Reproduced Constrained Policy Optimization (ICML 2017) baselines in MuJoCo from the official code (safe RL)

Selected Projects

•LLM-Guided Autonomous Robots *Mar. 2026 – Jun. 2026*

EE40078 Introduction to Multi-disciplinary Robotics, School of Electrical Engineering

– LLM-guided autonomous mobile delivery robot; RTAB-Map localization, path planning, and object pick-up
– Built the RGB-D grasping pipeline with a Kalman-filtered target estimate and the ROS mission orchestrator

•Hamiltonian Invariant-Driven Mode-Transition IMM for Satellite Maneuver Detection *Mar. 2026 – Present*

MO50125 The Principles and Applications of the Kalman Filter, Graduate School of Mobility

– IMM transition probabilities driven by a measurement-side CUSUM on the J_2-Hamiltonian residual
– Developed into RD-IMM [2], driving the transitions from the NIS of a reference coast filter outside the IMM to break the feedback loop of adaptive-TPM methods
– Detected small burns missed by a fixed-TPM IMM with few false declarations; cut post-burn position error by over 50% versus single filters

•PintOS-KAIST *Sep. 2024 – Dec. 2024*

CS330 Operating Systems, School of Computing

– Implemented kernel modules for process scheduling, syscall, memory management of an educational x86-64 OS

Honors and Awards

•Korea-U.S. Advanced Industries Youth Exchange Scholarship *Aug. 2025*

$9,000 for advanced-industry exchange study in the U.S.

Other Experience

•Republic of Korea Air Force (mandatory military service) *Apr. 2022 – Jan. 2024*

Information System Operator, Staff Sergeant, KAMD Operations Center Osan Air Base, KR

– Operated real-time data processing servers for the Korea Air and Missile Defense center; 24/7 availability

Technical Skills and Interests

Programming Languages: C, C++, Python
Libraries: ROS, NumPy, PyTorch
Miscellaneous tools: MATLAB, LaTeX, Docker
Languages: Korean (native), English (TOEFL iBT 5.5/6)
Extracurricular: STROKE (Tennis Club)